

Research article

The impact of economy policy uncertainty and oil price shocks on G20 banks' stock performance: Wavelet coherence and non-parametric causality in quantiles approach

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ABSTRACT

This study employs nonparametric causality-in-quantiles and wavelet coherence techniques to examine the impact of economic policy uncertainty and oil price variations on bank stocks in twelve prominent global economies. The results reveal that the effects of both economic policy uncertainty and oil prices on bank stock values vary significantly across countries and over time. Notably, during stress periods, we observe an inverse relationship between economic policy uncertainty and bank stocks in multiple countries, namely, Brazil, Canada, France, India, Russia, and the USA, with Japan exhibiting a particularly strong and long-term adverse correlation. Similarly, the influence of oil prices is primarily observed during crisis periods, but it demonstrates a substantial co-movement with bank stocks across the sample countries except Brazil. Our empirical analysis holds valuable implications for policymakers, bankers, investors, and portfolio managers.

1. Introduction

The importance of the banking industry is derived from its function as a financial intermediary, which includes providing employment opportunities, enabling capital movement, managing risks, supporting monetary policy, ensuring financial stability, and fostering economic growth. Both the general economy and the private sector benefit from these activities in terms of growth and stability.

In this study, we assess how banks as represented by banks' stocks (BANK) in twelve of the world's top economies responded to economy policy uncertainty (EPU) and shocks of the price of oil (OIL). By combining two distinct methodologies namely, nonparametric causality-in-quantiles and wavelet coherence analyses, our groundbreaking study adds this proof to the body of literature already in existence. In particular, we examine the influence of EPU and OIL on the returns and volatility of BANK in the selected countries. Our analysis employs nonparametric causality-in-quantiles to gauge these effects. Additionally, we assess the temporal dynamics of these impacts on BANK, accounting for both stable and unstable economic periods. Furthermore, we investigate variations across sample countries through wavelet coherence analysis.

Recognizing the inherent truth, bank managers and policymakers stand to gain significantly by comprehending the susceptibility of

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banks to external shocks such as EPU and OIL. This understanding is crucial for shaping policies geared towards stabilizing and safeguarding the performance of the banking sector. Moreover, portfolio managers and investors in bank stocks must grasp the dynamics of how external shocks affect bank stocks to effectively formulate hedging and investment strategies.

Economic Policy Uncertainty (EPU) is considered a risk factor, particularly in situations where future government regulations and policy frameworks are uncertain. EPU is characterized by its dynamic and rapidly changing nature. The associated market uncertainty can prompt both firms and individuals to defer purchases and investments, as highlighted by Ref. [1]. Notably, there is a strong interconnection between EPU and stock markets. For example [2], demonstrate that the Economic Policy Uncertainty in BRIC countries has a notable impact on stock markets. It is worth emphasizing that, to date, no prior study has comprehensively explored global evidence on how bank stocks respond to EPU. Addressing this research gap constitutes one of the primary objectives of this study.

The structure of this paper is as follows: Section 2 presents the material and methods. In Section 3, we present and analyze the empirical results. Finally, our conclusions are summarized in Section 4.

2. Literature review

The impact of EPU extends beyond capital markets to bank valuations, where prices decline as uncertainty increases [3]. Because of their slow or even negative growth rates and the significant level of uncertainty they face, banks have lower valuations. Uncertainty causes fresh financial friction that halts or at the very least slows economic expansion [4]. Demonstrate an inverse link between EPU and bank lending growth. Consequently, by tightening the restrictions on bank financing, uncertainty about future regulations can halt or reduce the increase of bank credit. On the same note [5], argue that Bank loans are a significant source of outside funding for the global real estate sector. By encouraging firm capital formation and household consumption, low cost bank lending can support real sector growth. Therefore, the cost of bank financing has a significant impact on the real estate sector's success. Besides, by degrading the overall external funding environment, uncertainty in government economic policy makes it harder for borrowers to meet their financial obligations. In a similar vein, company share prices typically decline. Their study reveals that interest rates on bank gross loans have a strong positive correlation with EPU. In their study [6], explore that EPU has a negative impact on the loan size by y shocking enterprises' demand for and banks' supply of credit resources, particularly in commercial banks in China. Eventually, this lowers the commercial banks' income.

Lastly, a relevant study by Ref. [7] this investigation delves into the intricate dynamics between Economic Policy Uncertainty (EPU) and the performance of banks in the context of India, casting a spotlight on the interplay with global crude oil prices. The analysis hinges on a dataset encompassing Indian commercial banks during the period spanning from 2000 to 2016. Within this exploration, a compelling narrative unfolds. A heightened EPU within India serves as a catalyst for a nuanced transformation in bank behaviour. Specifically, as Indian EPU experiences an uptick, banks respond by adopting a more pronounced risk-taking stance. This heightened risk-taking is particularly notable across dimensions such as insolvency risk, credit risk, and liquidity risk. Paradoxically, amidst this risk-inciting environment, banks also highlight an improvement in cost management efficiency and profitability, revealing an intriguing juxtaposition of effects. Moreover, this intricate relationship between Indian EPU and bank risk-taking is further amplified by the influence of international crude oil prices. This influence persists independently of the prevailing inflationary conditions within the country. The linkage underscores the globalized nature of the economic landscape and the intricate web of interconnected factors influencing banking behaviour. Unveiling yet another layer of complexity, the concentration levels within the banking sector contribute to shaping the risk profile of banks. Reduced concentration levels correlate with higher instances of insolvency and liquidity risk, coupled with a reduction in credit risk, cost efficiency, and profitability. Additionally, the annual expansion of broad money supply exhibits a noteworthy impact. This expansion corresponds to a decline in credit risk, while concurrently boosting cost efficiency and profitability.

In view of the pivotal role that banks assume within the economic framework, the culmination of these findings presents an intriguing outlook. Drawing on the synthesis of observations and insights, an important anticipation arises: a substantial inverse relationship is expected between Economic Policy Uncertainty (EPU) and the performance of banks (BANK).

[8] A comprehensive research endeavour unfolded, unearthing a multifaceted relationship between oil prices and the stock market. This intricate link was far from static, showcasing considerable variability across diverse industrial sectors and evolving over temporal horizons. Similarly [9], embarked on a parallel inquiry, venturing into the realm of oil price dynamics and their ripple effects on stock price indices. Their findings provided intriguing insights: changes in oil prices, whether ascending or descending, wielded an asymmetric influence on the stock price index within four distinct financial markets. However, the degree of this asymmetry wasn't uniform across all the markets scrutinized. The nuanced character of this asymmetry manifested uniquely, leading to varying impacts within the financial markets under scrutiny. Noteworthy among these insights was the notable distinction in the asymmetric effects of oil price volatility between China's financial market and its global counterparts. This observation underscores the uniqueness of China's economic landscape and the specific dynamics at play within its financial sector. Furthermore, the intricate dance between oil price volatility and Chinese stock price indices revealed intriguing patterns, unveiling layers of complexity inherent to this relationship. In essence, these investigations collectively emphasize that the relationship between oil prices and stock markets is a mosaic of variables, evolving over time and presenting distinctive characteristics across different economic landscapes. This realization urges us to delve deeper, untangling the intricate web of factors contributing to these dynamics, and forging a more comprehensive understanding of the global economic interplay.

The banking sector, being a critical component of the economy, is expected to be highly susceptible to oil price shocks. The repercussions of oil price fluctuations have been documented and thoroughly examined in the relevant academic literature. Economists

have yet to substantiate the mechanisms through which oil price shocks exert an influence on bank profitability, as there remains a dearth of econometric evidence in this regard [10,11] through their comprehensive investigation encompassing twenty-six nations, it has been established that the oscillations in oil prices yield a profound and substantive impact on the holistic macroeconomic landscape. This influence extends to intricate realms such as inflation dynamics and unemployment levels, specifically within the unique purview of OECD (Organization for Economic Cooperation and Development) countries. A study conducted by Ref. [12] delved into a distinctive dimension of this complex relationship. Their examination honed in on the repercussions of oil price fluctuations on bank performance, gauged through the lens of CAMEL indicators (Capital adequacy, Asset quality, Management, Earnings, and Liquidity). Spanning from 2000 to 2014, this study meticulously considered the potential interplay with country risk. The findings unveiled a compelling narrative: oil prices wield a discernible influence on banking performance. Heightened oil prices trigger a cascade of effects, leading to a reduction in various facets of banking prowess, spanning capitalization, management efficiency, earning capacity, and liquidity. This intricate interplay is, however, tempered by the stability of the country, with a notable emphasis on economic and political stability. In essence, the degree of stability exhibited by the nation serves as a mitigating force, offsetting the adverse impacts of oil price fluctuations on the banking sector's performance. The dynamics of bank profitability are subject to the influence of oil prices, wherein both direct and indirect mechanisms come into play. For oil-importing countries, the direct impact is characterized by a decline in lending activities and business operations within the oil-related sector, triggered by heightened oil prices. In contrast, the indirect effects materialize through macroeconomic avenues, encompassing fluctuations in inflation levels and reduced overall economic activities. The intricate and multifaceted nature of these pathways emphasizes the imperative for a comprehensive comprehension of the link between oil prices and bank profitability to effectively address their potential implications [13–15].

Concerning the Chinese economy, a study by Ref. [12] spanning the period from 2000 to 2014, encompassing 182 financial institutions, revealed a significant correlation between rising WTI (West Texas Intermediate) oil prices and diminished banking performance across key indicators such as capitalization, management efficiency, earnings power, and liquidity. The findings of the study by Ref. [14] provide compelling empirical evidence, substantiating that fluctuations in oil prices have a significant influence on the profitability of the Turkish banking sector. This influence operates through an indirect mechanism, mediated by the impact of oil price changes on inflation. Furthermore, the research uncovers a direct and adverse relationship between oil prices and bank profitability, primarily attributed to a reduction in lending activities within oil-related enterprises.

Nonetheless, the adverse ramifications can be ameliorated through the existence of economic and political stability. It is noteworthy that China, as a nation reliant on oil imports, demonstrates a multifaceted association with oil shocks, as substantiated by extant scholarly works. These studies underscore that the magnitude and orientation of such shocks are subject to variation contingent upon a country's status as an oil-exporting or oil-importing entity [16]. Empirical evidence by Ref. [13] concerning the impact of oil price fluctuations on a targeted cohort of oil-exporting countries. The research involves a comprehensive dataset encompassing 145 banks across 11 nations in the Middle East and North Africa (MENA) region, with observations spanning from 1994 to 2008. The study's results substantiate a positive association between oil price shocks and the profitability of the banking sector, irrespective of the particular criteria employed to delineate occurrences of oil price shocks. Hence, it is anticipated that there will be a negative correlation between BANK and EPU, especially during periods of stress [17].

Consequently, the existing body of literature within this realm underscores the pivotal role that oil shocks assume for both nations reliant on oil exports and those dependent on oil imports. Nonetheless, the potential directional influence of these oil shocks tends to deviate significantly between these two discernible categories.

Moreover [18], argue that a decrease in the price of oil creates additional difficulties for oil companies in fulfilling their debt obligations, potentially leading to adverse consequences for the asset quality of US banks. However, if banks were to cease lending to the Oil and Gas industries, it would inevitably result in turbulence within the oil market due to the fragile financial state of the US shale industry. This research investigation delves into the intricate dynamics of volatility transmission between the price movements of oil and the stock prices of the four leading American banks deeply intertwined with the oil sector. This intricate relationship is exemplified by the substantial and noteworthy volatility reactions observed in oil prices in response to perturbations in the stock prices of the banks. Similarly, the stock prices of these banks demonstrate significant volatility shifts as a direct result of shocks originating from oil price movements. This dual-sided phenomenon underscores the intricate interdependence between the oil industry and these key banks, with volatility acting as a prominent conduit through which these interactions manifest. On the same note [19], observed that during oil price boom episodes, credit infection gets worse, banking efficiency suffers, and the likelihood of default rises. These results support the existence of the resource curse and the notion that nations that depend too much on natural resources typically have less developed financial systems. Therefore, there is an expectation of a positive correlation between BANK and OIL, particularly during stress periods [20].

It's important to highlight that, to date, no prior study has delved into the worldwide repercussions of oil on the stocks of banks. Additionally, there has been a notable absence of research exploring how the relationship between oil and banks might evolve over time, particularly in the context of the unprecedented circumstances brought about by the COVID-19 outbreak. Consequently, the secondary goal of this study is to bridge this research gap, shedding light on these intricate dynamics and providing valuable insights into the nuanced impact of oil on banks, especially in the backdrop of the ongoing global pandemic.

3. Materials and methods

3.1. Data

This study analyses banks' exposure to two exogenous factors: oil price shocks (OIL) and economy policy uncertainty (EPU). OIL

data is sourced from Federal Reserve Economic Data (FRED) at <https://fred.stlouisfed.org/>, while EPU data is obtained from www.policyuncertainty.com. The bank stock index (BANK) represents all banks in each country and is fetched from www.investing.com. The study uses monthly data from the start of 1985 to the end of 2022, with variations in periods across countries due to the age of variables, particularly BANK and EPU. Furthermore, the sample includes twelve countries among G20 economies namely, Australia, Brazil, Canada, China, France, Germany, India, Japan, South Korea, Russia, UK, and the USA. Additionally, we converted the variables to their natural logarithm first difference computed as $\text{Ln}(P_{it} - P_{it-1}) * 100$

Where P_t is the monthly value of OIL and BANK and EPU for each country at month t .

4. Methodology

4.1. Nonparametric causality-in-quantiles

The nonparametric causality-in-quantiles method introduced by Ref. [21] is used to evaluate the relationship between bank stocks and economic policy uncertainty and oil prices. This strategy integrates the frameworks developed by Refs. [22,23]. This method was created with the aid of the nonparametric quantile causality framework by Ref. [22] and the k -th order nonparametric causality framework by Ref. [23]. In comparison to the traditional causality test, this methodology provides more benefits and aids in determining nonlinear causality [24–26]. Furthermore, this method is resistant to the extreme values in the data and captures the broad nonlinear dynamic dependencies. Bank stock price are represented by the dependent variable y_t while X_t represents economic policy uncertainty or oil prices. Following [22] we define the that the quantile-based causality as X_t does not cause y_t in the θ quantile with respect to the lag-vector of $\{y_{t-1}, \dots, y_{t-p}, X_{t-1}, \dots, X_{t-p}\}$, if:

$$Q\theta = y_t | y_{t-1}, \dots, y_{t-p}, X_{t-1}, \dots, X_{t-p} = Q\theta(y_t | y_{t-1}, \dots, y_{t-p}), \quad (1)$$

In Eq. (1), X_t Probably causes y_t in the θ -quantile with respect to $\{y_{t-1}, \dots, y_{t-p}, X_{t-1}, \dots, X_{t-p}\}$, if:

$$Q\theta = y_t | y_{t-1}, \dots, y_{t-p}, X_{t-1}, \dots, X_{t-p} \neq Q\theta(y_t | y_{t-1}, \dots, y_{t-p}), \quad (2)$$

In Eq. (2), $Q\theta = (y_t)$ represents the θ -th quantile of y_t which is dependent on t , and the quantiles are limited between 0 or 1 i.e., $0 < \theta < 1$.

To present the causality-in-quantiles test in a compressed manner, the following vectors,

$y_{t-1} \equiv (y_{t-1}, \dots, y_{t-p})$, $X_{t-1} \equiv (X_{t-1}, \dots, X_{t-p})$, $Z_t = (X_t, y_t)$, are defined.

The notion of conditional distribution plays a pivotal role in our analysis. We define two specific conditional distribution functions: $F_{y_t|Z_{t-1}}(y_t|Z_{t-1})$, and $F_{y_t|Z_{t-1}}(y_t|y_{t-1})$. These distributions encapsulate the probability distribution functions of y_t under the conditions of vectors Z_{t-1} and y_{t-1} , respectively. The conditional distribution vector $F_{y_t|Z_{t-1}}(y_t|Z_{t-1})$, exhibits the characteristic of being a continuous function of y_t for nearly all values within the domain of Z_{t-1} .

To simplify our notation, let's introduce the following notations: $Q\theta(Z_{t-1})$ as $Q\theta(y_t|Z_{t-1})$, and $Q\theta(Z_{t-1})$ as $Q\theta(y_t|Z_{t-1})$. Remarkably, a foundational outcome emerges: $F_{y_t|Z_{t-1}}\{Q\theta(Z_{t-1})|Z_{t-1}\}$ equates to θ , indicating a probability of 1. This intriguing result is derived directly from the principles elucidated in Equations (1) and (2).

Building upon the insights gleaned from the equations, we are poised to formulate the subsequent hypotheses that constitute the framework for performing a causality-in-quantiles test:

4.2. Hypotheses

- 1 The conditional distribution $F_{y_t|Z_{t-1}}(y_t|Z_{t-1})$ demonstrates a consistent and significant impact of Z_{t-1} on y_t across various quantiles.

$$H_0 : P\{F_{y_t|Z_{t-1}}\{Q\theta(y_{t-1})|Z_{t-1}\} = \theta\} = 1, \quad (3)$$

- 2 The conditional distribution $F_{y_t|Z_{t-1}}(y_t|Z_{t-1})$ reveals a persistent and notable influence of Z_{t-1} on y_t across different quantiles.

$$H_1 : P\{F_{y_t|Z_{t-1}}\{Q\theta(y_{t-1})|Z_{t-1}\} = \theta\} < 1, \quad (4)$$

These hypotheses form the bedrock upon which the causality-in-quantiles test is built, allowing us to examine the causal relationships between variables in a nuanced and quantile-specific manner.

In the pursuit of offering a more tangible comprehension of the practical implementation of causality-in-quantiles tests [22], introduced a distance measure that proves illuminating. This measure, denoted as J serves as a pivotal tool for quantifying the extent of causality across different quantiles. To delve into its intricacies, let us break down its components.

The foundation of this distance measure lies in the concept of the error term, symbolized as ε_t this term encapsulates the discrepancies between actual observations and expected values, playing a crucial role in our analysis. Additionally, $F_Z(Z_{t-1})$ represents the marginal density function of (Z_{t-1}) , providing insights into its probability distribution.

The driving force behind ε_t is intricately linked to the null hypothesis as formulated in Equation (3). This null hypothesis assumes a specific scenario where either $E[1\{y_t \leq Q\theta(y_{t-1})|Z_{t-1}\}] = \theta$, holds true, or it can equivalently be expressed as $1\{y_t \leq Q(y_{t-1})\} = \theta + \varepsilon_t$, denotes the indicator function, signaling whether a given condition is satisfied or not.

Bringing these elements together, the distance measure, as elucidated by Ref. [22], can be meticulously defined as follows:

$$J = E \left[\{F_{Y_t|Z_{t-1}}\{Q\theta(y_{t-1})|Z_{t-1}\} - \theta\}^2 F_Z(Z_{t-1}) \right], \quad (5)$$

In this equation, $F_{Y_t|Z_{t-1}}\{Q\theta(y_{t-1})|Z_{t-1}\}$ embodies the conditional distribution of y_t given Z_{t-1} , providing a glimpse into the relationship between past and current observations. The term θ serves as a reference point, against which the actual distribution is compared.

The square of the difference $F_{Y_t|Z_{t-1}}\{Q\theta(y_{t-1})|Z_{t-1}\} - \theta$ quantifies the deviation between the observed and reference distributions. This quantification is then weighted by $F_Z(Z_{t-1})$, reflecting the significance of different contexts represented by Z_{t-1} .

In essence, this distance measure J serves as a powerful metric to gauge the extent to which the null hypothesis holds across different quantiles, shedding light on the intricate causal dynamics within the data.

It's imperative to grasp the fundamental insights underlying Equation (3). Within this equation, the distance measure J is inherently constrained by non-negativity; that is, $J \geq 0$. The crucial point to comprehend is that the equality $J = 0$ is realized if and only if the null hypothesis H_0 stated in Equation (5) holds true. In contrast, when considering the alternative hypothesis in equation F(4) $J > 0$ becomes evident.

In Equation (5), the practical implications of this concept are elucidated further by Ref. [22]. This equation outlines the kernel-based sample analog of J , tailored to real-world application scenarios. The definition of this analog, denoted as $\hat{J}_T$, involves several key elements that warrant careful consideration.

$$\hat{J}_T = \frac{1}{T(T-1)h^{2p}} \sum_{t=p+1}^T \sum_{s \neq p+1, s \neq t}^T K\left(\frac{Z_{t-1} - Z_{s-1}}{h}\right) \hat{\varepsilon}_t \hat{\varepsilon}_s, \quad (6)$$

Essentially, this analog $\hat{J}_T$ calculates the weighted sum of the cross-products of estimated error terms using the kernel function. The kernel function accounts for the influence of the context variables Z_{t-1} and Z_{s-1} through the smoothing parameter h . This measure encapsulates the degree of deviation from the null hypothesis within the data, providing a quantitative assessment of causal relationships across different quantiles.

In conclusion, Equation (3) unveils the foundation for quantifying the relationship between the null hypothesis H_0 and the distance measure J . The subsequent development of the analog $\hat{J}_T$ offers a practical means of assessing causality-in-quantiles tests through a meticulously designed calculation incorporating the various components of the data and its context.

Let us delve into Equation (6), a cornerstone in our understanding. Within this equation, $K(\cdot)$ embodies the kernel function, h symbolizes the bandwidth, t pertains to the sample size, and p denotes the lag-order. A fundamental element we need to grasp is that $\hat{\varepsilon}_t$, the estimated regression error term, plays a pivotal role. This term is illuminated by the following equation:

$$\hat{\varepsilon}_t = 1\{y_t \leq \hat{Q}\theta(y_{t-1})\} - \theta, \quad (7)$$

Here, $\hat{Q}\theta(y_{t-1})$ stands as an estimate of the θ -th conditional quantile of y_t given y_{t-1} . This quantile estimation is paramount, as it uncovers the underlying behavior of y_t based on its history. Importantly, the essence of Equation (7) lies in its capacity to assess deviations from the baseline θ due to contextual variations.

To elaborate on the estimation of $\hat{Q}\theta(y_{t-1})$, we turn our attention to Equation (8):

$$\hat{Q}\theta(y_{t-1}) = \hat{F}_{y_t|y_{t-1}}^{-1}(\theta y_{t-1}), \quad (8)$$

Here, $\hat{F}_{y_t|y_{t-1}}(y_t, y_{t-1})$ encapsulates the Nadaraya-Watson kernel estimator, a cornerstone in non-parametric estimation techniques. This estimator characterizes the relationship between y_t and y_{t-1} within the context of quantiles. The inverse function applied to θ and y_{t-1} yields an estimate of the quantile.

In essence, the Nadaraya-Watson kernel estimator accounts for the influence of past observations on current quantiles by employing the kernel method. This technique allows us to gauge the impact of y_{t-1} on the quantile estimation for y_t .

To summarize, Equation (6) provides us with a means to examine quantile-based relationships with the support of the kernel function and bandwidth. Equation (7) helps us understand how the estimated error terms manifest deviations from the baseline assumption. Equation (8) introduces us to the heart of the quantile estimation process, involving the Nadaraya-Watson kernel estimator to unravel the intricate dynamics between past and current observations. Together, these equations pave the way for comprehensive analysis of causal relationships within quantile contexts.

$$\hat{F}_{y_t|y_{t-1}}(y_t, y_{t-1}) = \frac{\sum_{s=p+1, s \neq t}^T L\left(\frac{y_{t-1} - y_{s-1}}{h}\right) 1(y_s \leq y_t)}{\sum_{s=p+1, s \neq t}^T L\left(\frac{y_{t-1} - y_{s-1}}{h}\right)}, \quad (9)$$

Equation (9) encompasses a significant aspect of our analysis, In this equation:

L represents the kernel function, an integral component in estimating relationships.

h denotes the bandwidth parameter, crucial for determining the extent of smoothing applied to the kernel.

The summations are carried out over all s except t , with $p+1 \leq s \leq T$. This encompasses a range of observations, reflecting the impact of various contexts on the estimation.

$1(y_s \leq y_t)$ serves as an indicator function, evaluating whether y_s is less than or equal to y_t .

Moving forward, our endeavour involves scrutinizing the direction of causality, whether it flows from economic policy uncertainty or oil prices to bank stock prices, or vice versa. Notably, causality in variance alludes to the transmission of volatility, a concept that can persist even in the absence of mean causality (1st moment). Building upon the groundwork laid by Ref. [22], we extend our analysis to encompass the causality test for the 2nd moment.

However, unravelling causality in higher-order moments introduces complexities. It's paramount to approach this test carefully, as the rejection of causality in the m -th moment doesn't imply the absence of causality in the k -th moment for $m < k$. To navigate this intricate terrain, we employ the nonparametric Granger quantile causality technique pioneered by Ref. [23]. This approach proves valuable in assessing the presence of causality in moments beyond the mean.

Our analysis unfolds with a meticulous procedure, beginning by examining causality for y_t through the equation:

$$1\{y_t \leq \hat{Q}\theta(y_{t-1})\} = \theta + \varepsilon_t$$

This equation serves as a foundational step, allowing us to discern the causal relationships within the higher moments, paving the way for comprehensive insights into the intricate dynamics of the variables under consideration.

$$y_t = g(y_{t-1}) + \sigma(X_{t-1})\varepsilon_t, \quad (10)$$

We now turn our attention to the intricacies of the variables at play. In this context, ε_t represents a white noise process, while $\sigma(\cdot)$, and $g(\cdot)$, represent functions with unknown properties. It's important to note that these functions adhere to the stationary properties of y_t . It's worth mentioning that this illustration focuses on the predictive power of X_{t-1} to y_t^2 , rather than capturing linear or nonlinear causal relationships from X_{t-1} to y_t , especially when $\sigma(\cdot)$ embodies a generalized nonlinear function. Notably, Equation (10) elucidates that the incorporation of the squared term of X_{t-1} into the nonlinear function $\sigma(\cdot)$ is not a compulsory consideration.

With this understanding in mind, we transform Equation (10) into two distinct forms: Equations (11) and (12). These equations encapsulate the null and alternative hypotheses, respectively, in the context of causality in variance. These hypotheses allow us to quantitatively assess the presence or absence of causal relationships within the moments of interest.

For the null hypothesis, we express:

$$H_0 : P\{Fy_t^2 | Z_{t-1}\{Q\theta(y_{t-1}) | Z_{t-1}\} = \theta\} = 1, \quad (11)$$

This hypothesis posits that the probability of the conditional distribution $Fy_t^2 | Z_{t-1}$ evaluated at $Q\theta(y_{t-1})$ given Z_{t-1} being equal to θ is unity.

In contrast, the alternative hypothesis is formulated as:

$$H_1 : P\{Fy_t^2 | Z_{t-1}\{Q\theta(y_{t-1}) | Z_{t-1}\} = \theta\} < 1, \quad (12)$$

Under this alternative hypothesis, we explore the possibility that the aforementioned probability is less than one, indicating a departure from the null hypothesis. This provides the framework for assessing whether causal relationships in variance exist, shedding light on the complex dynamics between the variables involved. By delving into these equations - (10), (11), and (12) - we embark on a rigorous analytical journey to understand the variance-based causality within the context of nonlinear functions. These equations empower us to unveil the hidden relationships that influence the variability in the variables under consideration.

Proceed to derive a practical test statistic based on the insights from Equation (10). To do this, we incorporate the concept of y_t^2 into Equations (6)–(9). This adjustment allows us to harness the [22] approach and address the challenge associated with causality in the conditional first moment (mean). Overcoming this challenge becomes instrumental in exploring causality in the second moment (variance), while also enabling us to extend our analysis to higher moments. This extension is grounded in the following model:

$$y_t = g(X_{t-1}, y_{t-1}) + \varepsilon_t, \quad (13)$$

Here, Equation (13) defines the interplay between y_t , X_{t-1} , y_{t-1} , and the error term ε_t . This framework accommodates both mean causality and the implications for variance causality, serving as a cornerstone for exploring higher moments.

With this framework established, we delve into the concept of causality in higher-order quantiles. This is encapsulated by the null and alternative hypotheses:

Null Hypothesis:

$$H_0 : P\{Fy_t^k | Z_{t-1}\{Q\theta(y_{t-1}) | Z_{t-1}\} = \theta\} = 1 \text{ for } k = 1, 2, \dots, K, \quad (14)$$

This hypothesis posits that the probability of the conditional distribution $Fy_t^k | Z_{t-1}$ evaluated at $Q\theta(y_{t-1})$ given Z_{t-1} being equal to θ is unity for all k up to K .

$$H_1 : P\{Fy_t^k | Z_{t-1}\{Q\theta(y_{t-1}) | Z_{t-1}\} = \theta\} < 1 \text{ for } k = 1, 2, \dots, K, \quad (15)$$

Under the alternative hypothesis, we explore the possibility that the aforementioned probability is less than one, implying deviations from the null hypothesis for each k within the range up to K . This intricate framework allows us to unravel the presence of causality in higher moments, thus providing deeper insights into the underlying relationships among the variables. In essence, these equations - (10), (13), (14), and (15) - empower us to construct a feasible test statistic that considers squared terms, overcome challenges in causality, and extend our exploration to encompass higher moments. This analytical foundation enhances our capacity to

understand and interpret the nuanced relationships that govern the interactions among the variables in question.

To encapsulate the entirety of this conceptual framework, we assert that x_t Granger causes y_t in the θ -quantile, extending this influence up to the k -th moment, as enabled by Equation (14). This premise drives us to construct the test statistic introduced in Equation (6) for each k . However, the complexity emerges when endeavoring to integrate the diverse statistics for every $k = 1, 2, \dots, K$ as these statistics demonstrate joint correlations as highlighted by Ref. [23].

To address this intricate challenge, we draw upon the sequential testing technique pioneered by Ref. [23], making a few tailored adjustments. The initial step involves examining nonparametric Granger causality in the first moment, with $K = 1$. It's vital to note that rejecting the null hypothesis of non-causality in this context doesn't imply the absence of causality in the second moment. However, such a rejection in the first moment serves as a robust indicator of the presence of Granger quantile causalities in variance.

Building upon this foundation, we extend our testing approach by setting $K = 2$. This choice allows us to explore the existence of both causality in variance and causality in variance-invariance successively. This two-tiered strategy enhances our analytical insight, capturing diverse dimensions of causal relationships.

The empirical application of this quantile-based causality testing framework involves making three pivotal selections:

The choice of p (lag order) is grounded in the Schwarz Information Criterion (SIC). We adopt a lag order of 1, based on the VAR model encompassing bank stock prices, economic policy uncertainty, and oil prices. The SIC is preferred for its parsimony, providing a prudent selection of lags in comparison to other alternatives. Additionally, the SIC effectively mitigates the issues of over parameterization often encountered in other nonparametric methodologies.

Determining the bandwidth (h) is accomplished using the least squares cross-validation method. This choice ensures an optimal bandwidth parameter, essential for precision in our analysis.

For the kernel functions $K(\cdot)$ and $L(\cdot)$ in Equations (6)–(9), we opt for Gaussian-type kernels. This selection aligns with the characteristics of the data, enhancing the validity of our quantile-based causality testing.

This comprehensive testing approach requires thoughtful considerations in selecting the lag order, bandwidth, and kernel type. It encompasses an integrated methodology that bridges causal relationships in different moments, providing a powerful framework to dissect intricate dynamics among the variables.

4.3. Wavelet coherence analysis

Our investigation takes a cue from the methodology laid out by Ref. [27], where we probe the relationship between economic policy uncertainty, oil prices, and bank stock prices through the lens of wavelet analysis. Central to this approach is the wavelet coherency (WSC), a tool that delves into the interconnections between time series data. This coherency is mathematically defined as follows:

$$WSC = R_{XY}^2(\tau, s) = \frac{|S(s^{-1}W_{XY}(\tau, s))|^2}{S(s^{-1}|W_X(\tau, s)|^2)S(s^{-1}|W_Y(\tau, s)|^2)} \quad (16)$$

S represents a smoothing operator that functions in both the time and scale domains.

W_{XY} stands for the cross-wavelet power (XWP), a metric that captures the local covariance between the two time series at each scale.

R_{XY}^2 is akin to a traditional correlation coefficient, bounded within the range $0 \leq R_{XY}^2(\tau, s) \leq 1$.

The fascinating aspect of the WSC is that it's confined to positive values, steering us toward the utilization of phase differences, φ_{XY} to ascertain lead-lag relationships:

$$\varphi_{XY} = \tan^{-1} \left(\frac{\text{Im}\{S(s^{-1}W_n^{XY}(s))\}}{\text{Re}\{S(s^{-1}W_n^{XY}(s))\}} \right) \text{ with } \varphi_{XY} \in [-\pi, \pi] \quad (17)$$

In this equation:

Im and Re refer to the imaginary and real parts of the smoothed XWP, respectively.

The phases φ_{XY} are visually represented by arrows on the WSC plots, acting as insightful indicators of the causal direction or lead-lag relationships between the two time series. When arrows point to the right (left) or left (right), it signifies a positive (negative) connection or in-phase (out-of-phase) movement between the time series. The presence of up-right and down-left pointing arrows suggests that the first time series is ahead of the second. Conversely, the absence of these orientations indicates the reverse relationship.

In essence, this wavelet-based analysis offers us a deeper insight into the dynamic interactions and causal relationships among economic policy uncertainty, oil prices, and bank stock prices. The WSC and associated phase differences not only quantify these connections but also visually illustrate the intricate lead-lag relationships, enriching our understanding of the underlying dynamics.

5. Results and discussion

5.1. Results of nonparametric causality-in-quantiles test

The outcomes stemming from the nonparametric causality-in-quantiles analysis for both Economic Policy Uncertainty (EPU) and

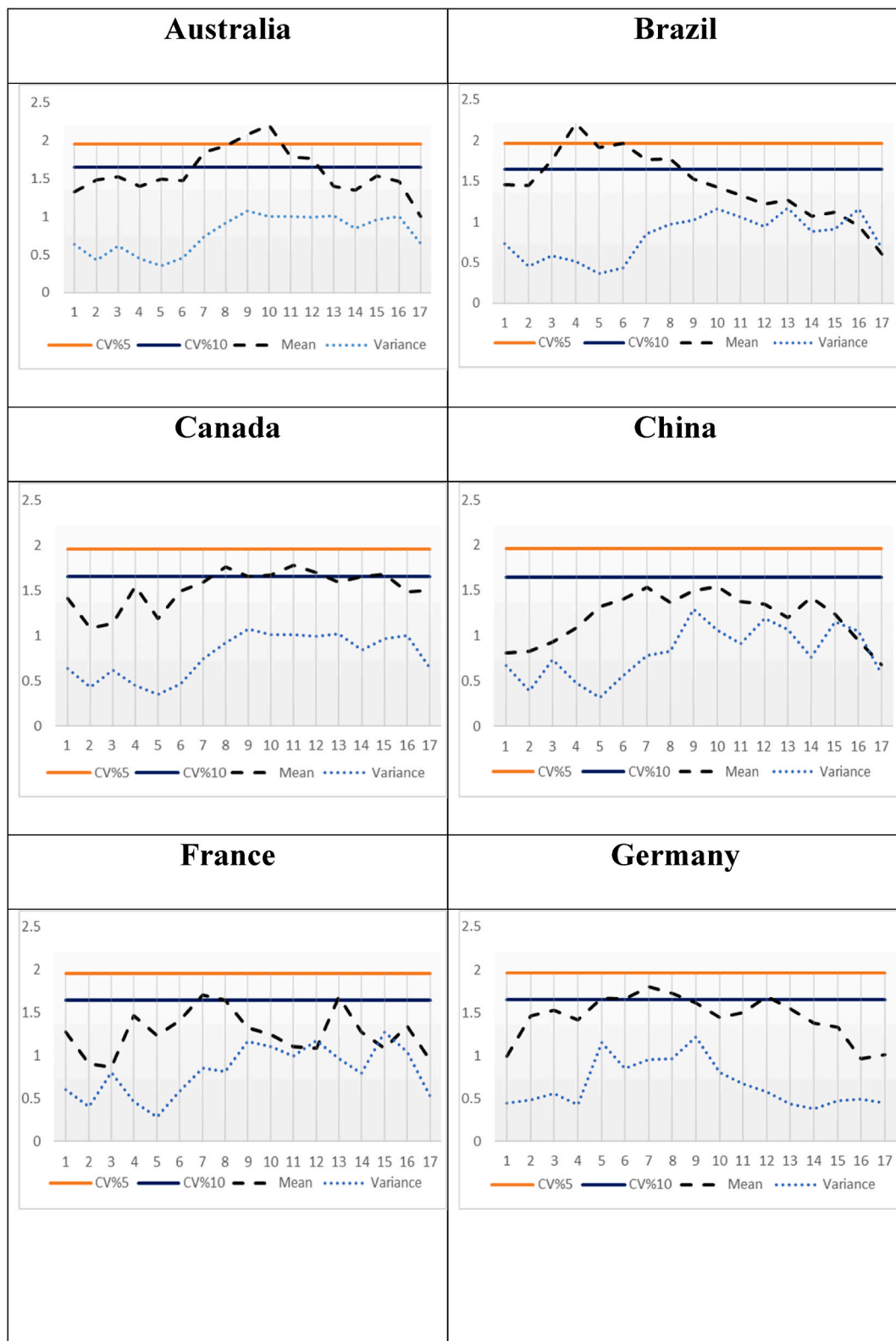


Fig. 1. The nonparametric causality-in-quantiles relationship from EPU to BANK stocks.

Note: The focal point of this exploration is the utilization of nonparametric causality test statistics. Notably, these statistics hold critical values of 1.96 and 1.65, aligning with the significance levels of 5% and 10%, respectively. Central to this investigation is the null hypothesis, a cornerstone in

assessing the intricate dynamics at play. In essence, this null hypothesis posits the absence of any causality originating from Economic Policy Uncertainty (EPU) to BANK stocks, serving as the foundation upon which the analytical journey is undertaken.

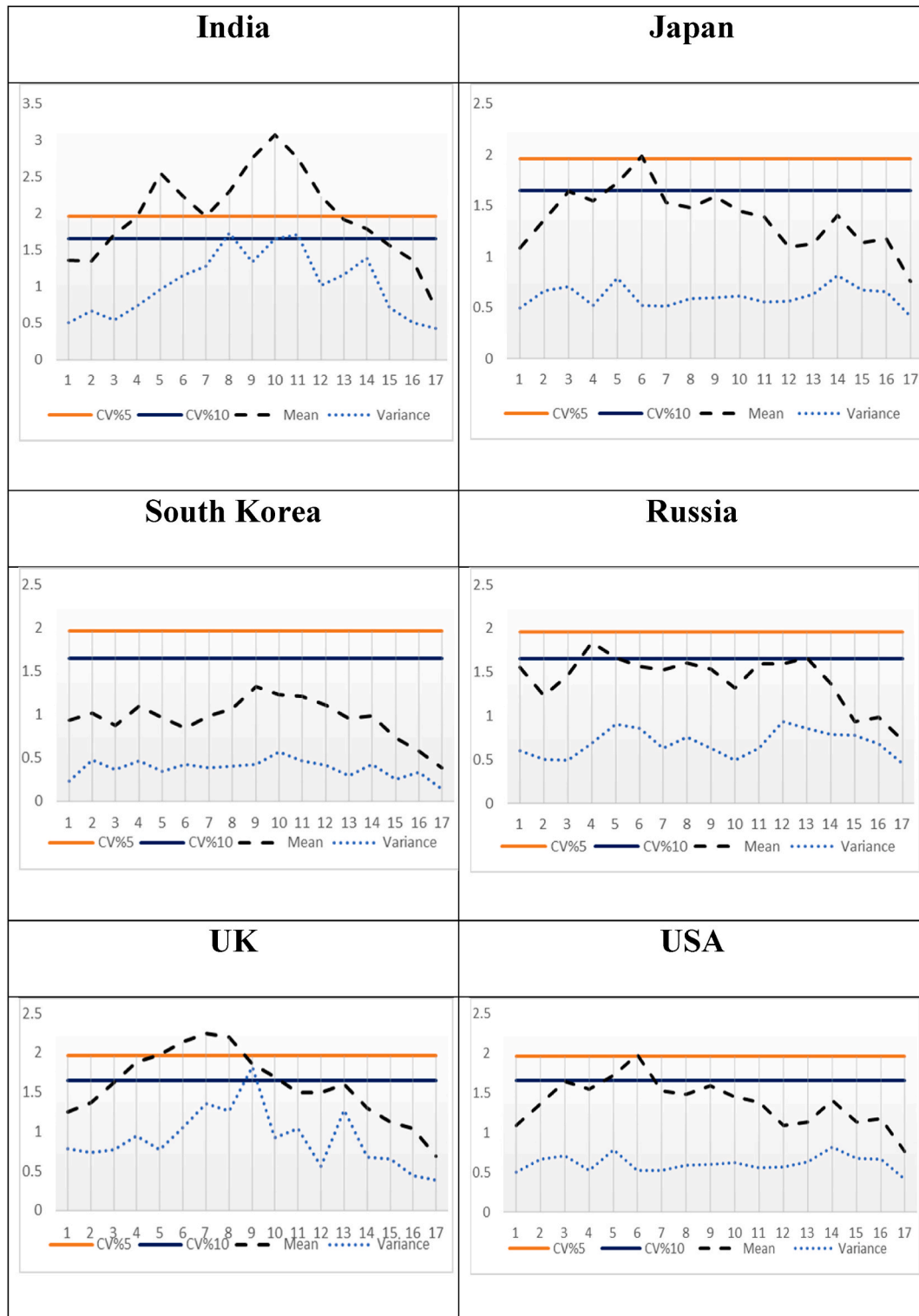


Fig. 1. (continued).

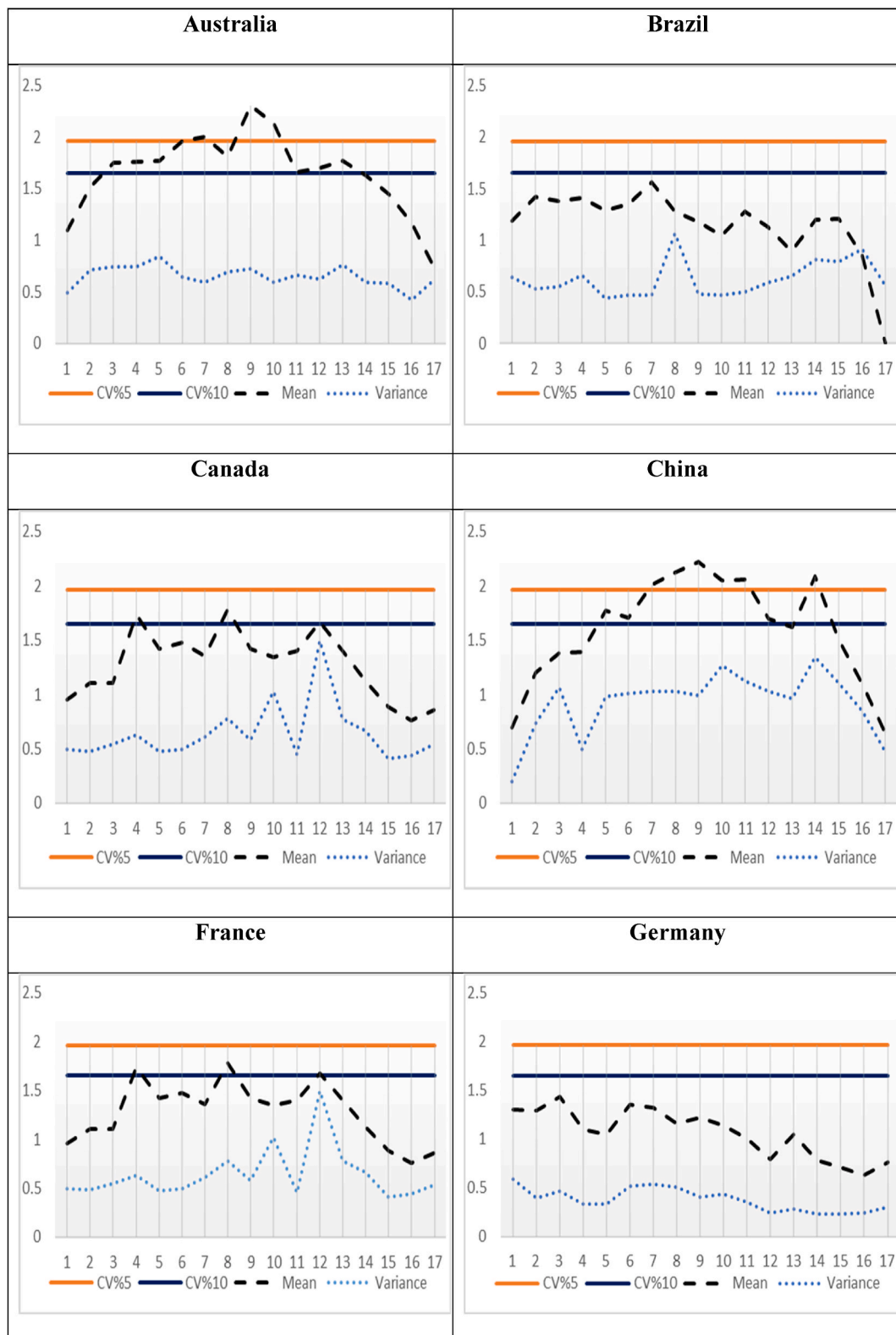


Fig. 2. The nonparametric causality-in-quantiles relationship from OIL to BANK stocks.

Note: The analysis centers on nonparametric causality test statistics, where the values of 1.96 and 1.65 hold significance as they correspond to the

5% and 10% levels of significance, respectively. Central to the investigation is the null hypothesis, a pivotal element in this exploration. This null hypothesis posits the absence of any causality stemming from OIL to BANK stocks, presenting a foundation upon which the inquiry is built.

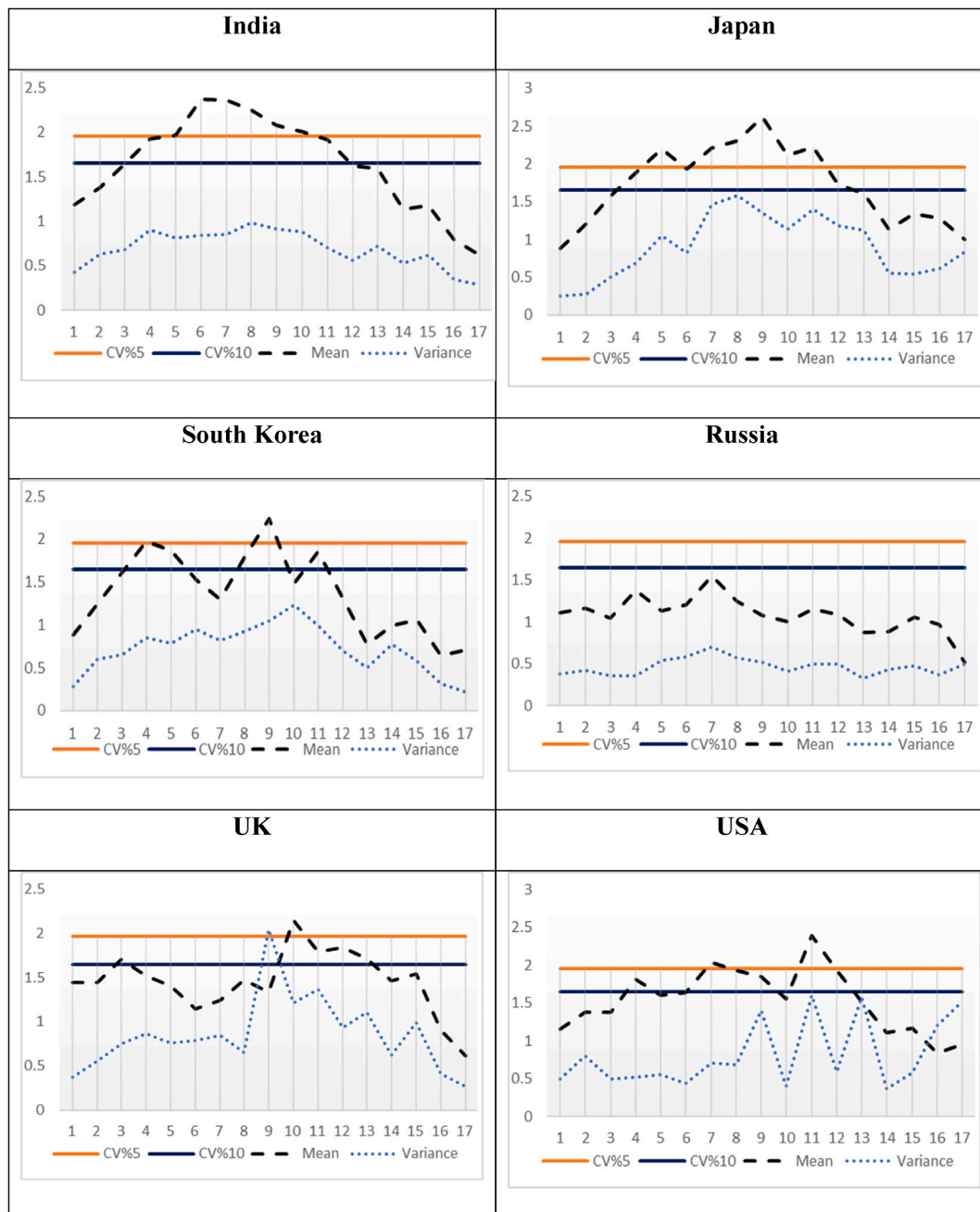


Fig. 2. (continued).

Oil Prices (OIL) influencing Bank Stock Prices (BANK) are meticulously presented in Figs. 1 and 2, respectively. In these visual representations, the horizontal axis portrays the quantiles under consideration, while the vertical axis serves as the stage for the nonparametric causality test statistics to unfold. Our exploration into the causal relationships takes a nuanced perspective by examining how variations in quantiles of EPU and OIL potentially impact BANK. These quantiles shed light on specific segments of the distributions, allowing us to pinpoint where causality might be prominent or subtle. These analyses are an essential facet of grasping the intricate interactions between these variables across different ranges of values.

Within the broader context of financial and economic dynamics, these figures offer insights that transcend mere numerical values. By visually depicting the nonparametric causality test statistics across quantiles, we gain a comprehensive understanding of the

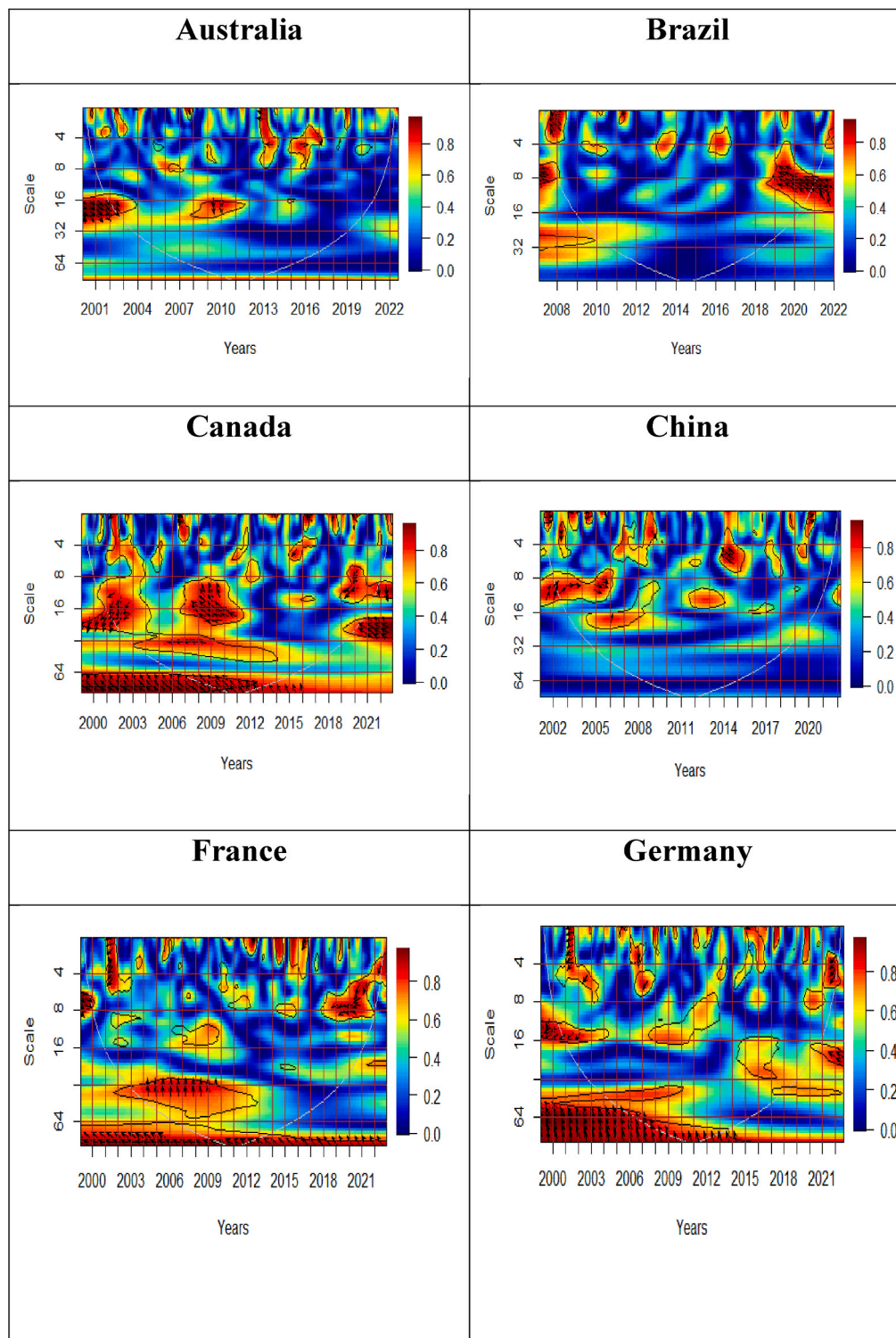


Fig. 3. Wavelet coherence between EPU and BANK stocks. *Note:* The contours, representing the 5% significance level against the backdrop of noise, are presented prominently, drawing attention to crucial thresholds. The power spectrum is artfully depicted through a color spectrum that transitions from blue (indicative of minimal or absent correlation) to red (indicating robust correlation), offering a comprehensive insight into the underlying dynamics. The directional interplay, intricately captured through the placement of arrows, serves as a guide to understanding the complex temporal relationships. When these arrows are oriented to the right or left, they signify the synchronization (in-phase) or disparity (out of phase) in the movement of the two series. Furthermore, these arrows hold additional meaning: a rightward arrow conveys a positive correlation,

while a leftward arrow denotes a negative correlation. The arrows that assume an up-right or down-left orientation assume a special significance. These arrows illuminate the role of EPU in leading the variations observed in BANK stocks. Conversely, when the arrows point in a different direction, it signifies the influence exerted by BANK stocks in shaping the fluctuations witnessed in EPU. This intricate interplay of arrows and color shades collectively offers an insightful panorama of the intricate temporal and directional associations governing the relationship between EPU and BANK. . (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

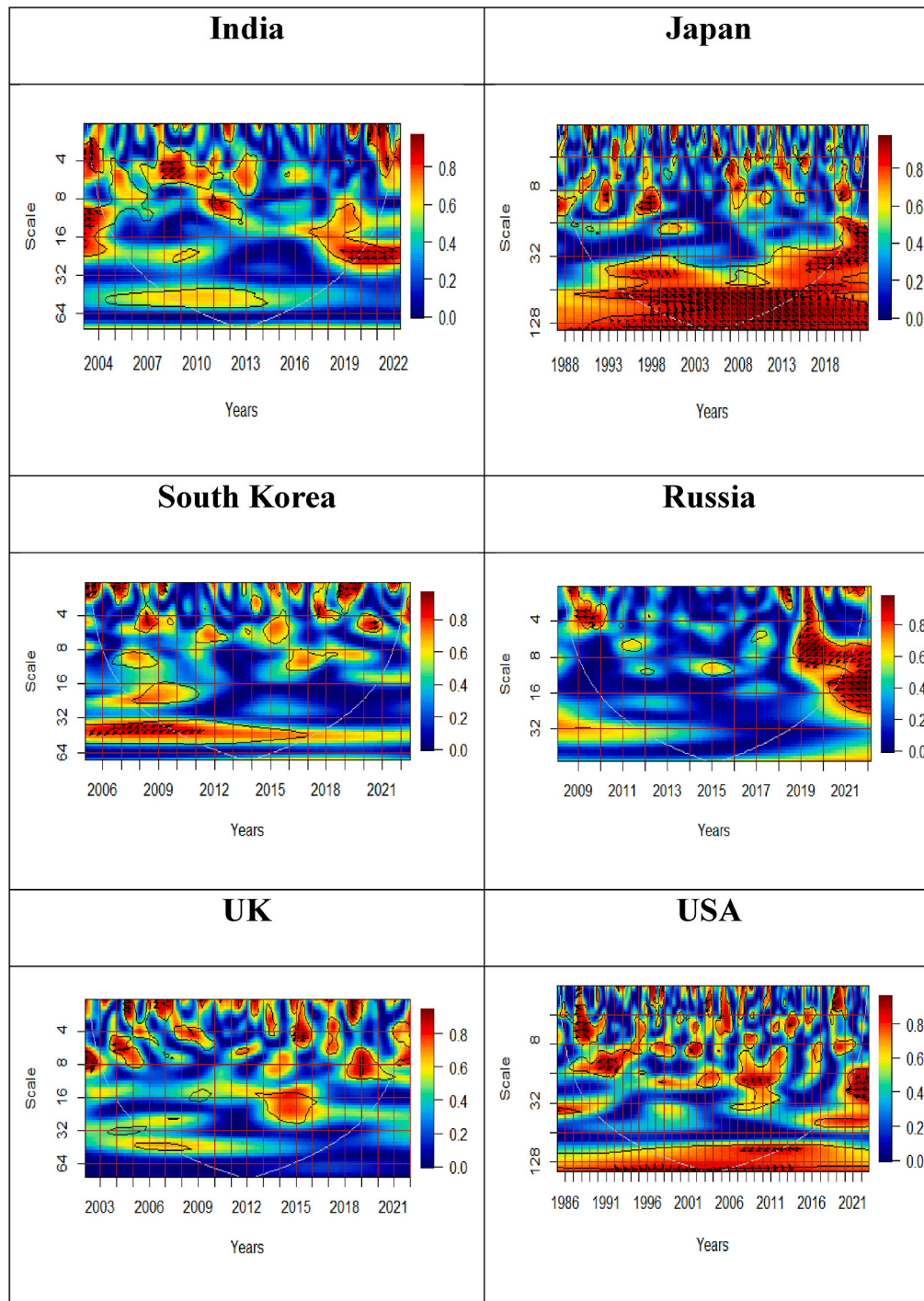


Fig. 3. (continued).

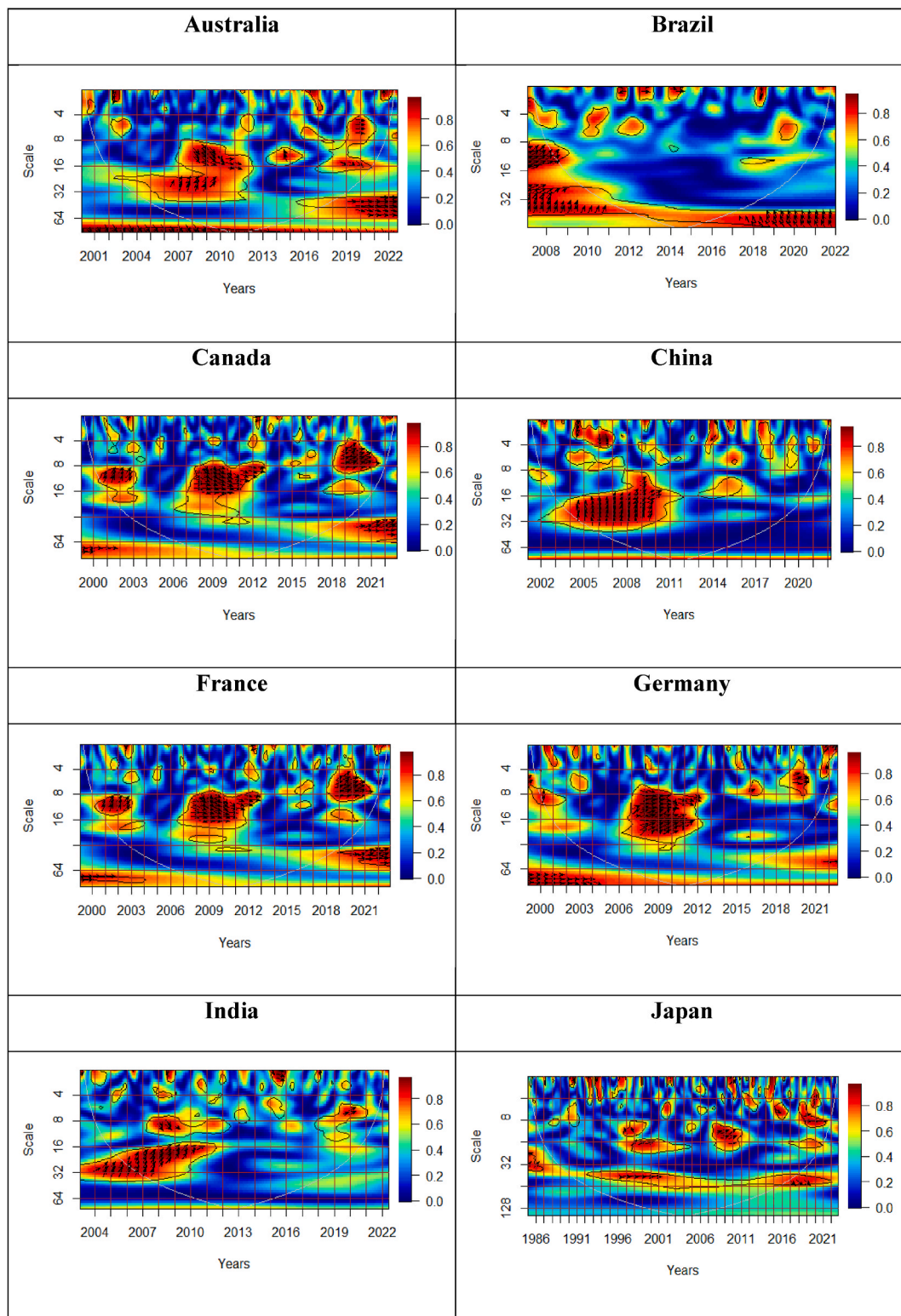


Fig. 4. Wavelet coherence between OIL and BANK stocks. *Note:* The contours depicting the 5% significance level against the backdrop of noise are prominently illustrated. The power spectrum, represented through colours ranging from blue (indicative of low or negligible correlation) to red (signifying strong correlation), contributes to a comprehensive visual understanding. The directional interplay between the two time series is effectively conveyed through the placement of arrows. When arrows are directed to the right or left, they denote the synchronization (in-phase) or

disparity (out of phase) in movement between the two series. Further nuances emerge when interpreting these arrows: a rightward arrow signifies positive correlation, while a leftward arrow indicates a negative correlation. The presence of up-right and down-left pointing arrows holds particular significance. Such orientation underscores the role of OIL in leading the variations observed in BANK stocks. Conversely, when these arrows assume a different alignment, it signals that it is the BANK stocks that are influencing and guiding the changes witnessed in OIL. This intricate web of arrows and colours weaves together a comprehensive narrative of the temporal and directional relationships governing the interplay between OIL and BANK. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

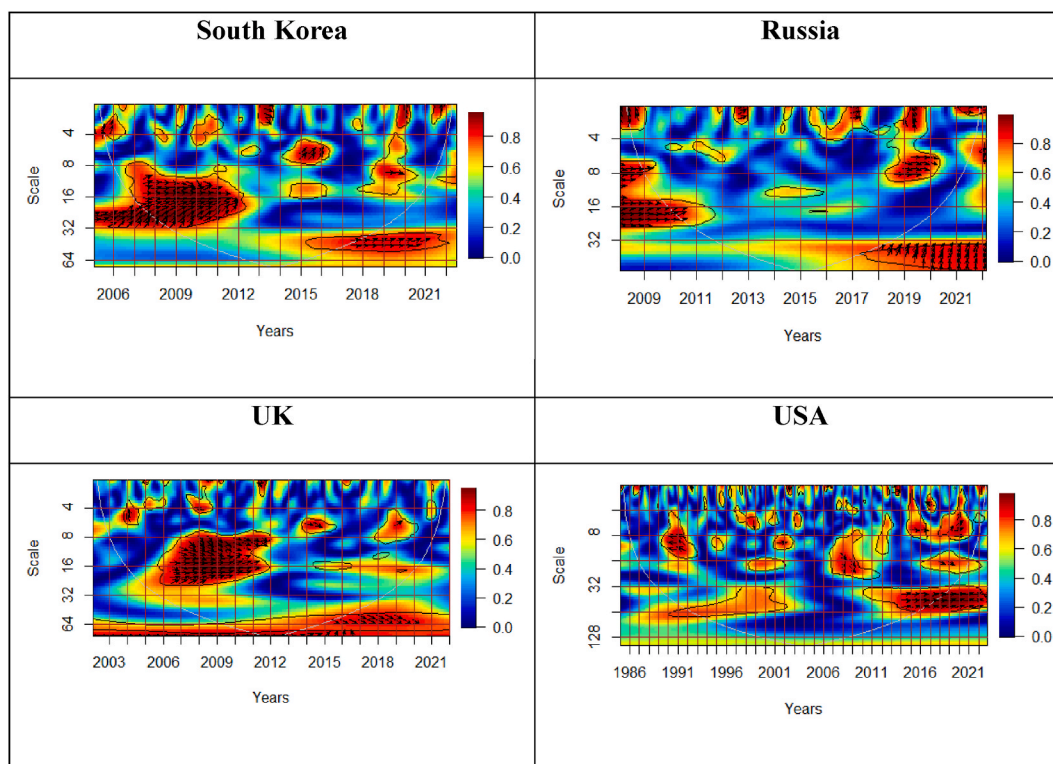


Fig. 4. (continued).

magnitude and nature of the causal relationships, enhancing our ability to draw meaningful conclusions about the influence of EPU and OIL on BANK. These graphical representations enrich our interpretation of the data, enabling us to uncover patterns, trends, and potential lead-lag relationships that may not be immediately apparent through purely numerical analysis. The threshold value at the 5% significance level is 1.96, while the pivotal value at the 10% significance level stands at 1.65. The horizontal lines in orange depict the critical value at the 5% level, while the deep blue lines correspond to the critical value at the 10% level. The outcomes showcased in Fig. 3 underscore a notable contrast in the quantile-based causality assessment between the examination of mean and variance. Additionally, these findings unveil substantial discrepancies across distinct nations.

The findings presented in Fig. 1 unveil intriguing insights into the causal relationships within the context of mean and variance. Across Canada, China, France, Germany, Japan, South Korea, Russia, and the USA, the analysis indicates the absence of quantile-based causality, both in terms of mean and variance, at a 5% significance level. This outcome is likely tied to the robustness of their banking sectors and the relatively steady economic policies pursued in these nations.

However, when turning our attention to India, a distinct pattern emerges. The quantile-based causality test concerning mean reveals that the null hypothesis is rejected at the 5% level of significance, specifically from the 4th quartile to the 13th quartile. This signals a notable relationship between Economic Policy Uncertainty (EPU) and banks, albeit within specific quantile ranges. Notably, this observation aligns with the findings of [7], who observed a similar trend in India where heightened economic policy uncertainty corresponded to increased risk-taking by Indian banks.

Turning to Australia, Brazil, and the UK, the analysis indicates statistically significant causality in mean only within distinct quartiles: the 2nd, 3rd, and 4th quartiles, respectively, at a 5% significance level. This suggests a comparatively weaker effect of EPU on banks in these countries, possibly confined to particular time periods, such as global financial crises or economic upheavals.

It's worth noting that these quantile-based findings serve as a precursor to a more nuanced exploration. The results of wavelet coherence tests are anticipated to offer additional layers of insight, enabling us to delve deeper into the complex interplay between economic policy uncertainty, bank behaviour, and economic conditions. This intriguing analysis is elaborated upon in the subsequent discussions.

Based on an extensive and advanced analysis, we have delved into the intricate relationship between the oil industry and the

banking sector across diverse countries. The outcomes, depicted in Fig. 2, provide compelling insights. Specifically, the analysis underscores that within Brazil, Canada, France, Germany, and Russia, no statistically significant quantile-based causality—whether in terms of mean or variance—exists from Oil (OIL) to Banks (BANKS) at a 5% significance level.

However, the narrative takes a different turn for Australia, China, India, Japan, South Korea, the UK, and the USA. In these countries, the null hypothesis of no quantile causality is challenged. In fact, this hypothesis is rejected across a spectrum of one to five quantiles. This intricate pattern unveils a discernible causal effect of oil on banks within these nations.

When probing the quantile-based causality in variance, a specific spotlight emerges. The impact is statistically significant only in the UK, particularly within the 9th quantile. This finding resonates with prior research by Refs. [18,19], both of which emphasized the intricate interplay between oil prices and the performance of the banking sector.

A pivotal realization stemming from these results is the undeniable influence of fluctuating oil prices on banks. Consequently, it becomes imperative for financial institutions to actively monitor and prudently manage their exposure to the energy sector. Such vigilance equips banks to adeptly navigate the challenges and seize the opportunities presented by dynamic oil price changes. Recognizing and mitigating the risks entailed in oil price volatility hold the key to sustaining stability and ensuring the profitability of banks that operate in countries where the causal link between oil and banking has been established.

6. Results of wavelet coherence analyses

Fig. 3 presents the impact of EPU on BANK across various sample countries. The findings closely align with the outcomes of nonparametric causality-in-quantiles tests. Specifically, no significant or apparent effect of EPU on BANK is observed in Australia, China, Germany, South Korea, and the UK. In Brazil, however, there is an inverse correlation between BANK and EPU, but only during the COVID-19 outbreak. Similarly, in Canada, France, Russia, and the USA, EPU is found to have a negative association with BANK during both the 2008 financial crisis and the COVID-19 outbreak, particularly in the moderate frequency band. It's important to note that this impact is short-term in India, occurring only during both crises. In Japan, a strong negative influence of EPU on BANK is observed over the long run, encompassing normal and stress periods. This finding supports the research [3], which also discovered an inverse relationship between EPU and bank valuation. An explanation for this is that the presence of uncertainty creates additional obstacles in financing that impede or at least hinder economic growth [4]. Demonstrate that Economic Policy Uncertainty (EPU) exhibits an inverse relationship with the credit growth of banks. As a result, uncertainties surrounding future policies can halt or reduce the growth of bank credit by imposing greater constraints on bank financing. Moreover, the strong negative correlation between EPU and BANK during crises can be attributed to negative market sentiment, increased volatility, reduced investor confidence, and hesitation in business investment. These factors collectively drive down stock prices, resulting in a strong and negative correlation between economic policy uncertainty and stock prices during times of crisis.

The exposure of BANK to OIL is reported in Fig. 4. A strong co-movement between OIL and BANK is observed over medium term period and across the sample countries. This effect is limited to the period of the 2008 financial crisis and/or the COVID-19 period. The only exception is Brazil, which shows no significant association with OIL. Oil prices have significant implications for financial markets, including the banking sector. A substantial drop in oil prices, specifically during financial crises can serve as an indicator of economic weakness or instability, resulting in market turbulence. This can have a ripple effect on the overall sentiment in financial markets, impacting investor confidence and risk appetite. As a result, banks may experience changes in their funding costs, liquidity conditions, and access to capital markets, as investor sentiment and risk aversion can influence the overall financial landscape.

7. Conclusion

This study examines the relationship between banks' performance, measured by their stock index, and the exposure to both economic policy uncertainty and oil price shocks in twelve prominent global economies. To analyze this relationship, we employ a nonparametric causality-in-quantiles approach and utilize wavelet coherence. The findings indicate that economy policy uncertainty has a predominantly negative impact on bank stocks across most countries, particularly during stress periods. Furthermore, we observe a strong positive correlation between oil prices and bank stocks during crises, except for Brazil. It is important to acknowledge that the analysis is constrained by the limited availability of data, particularly concerning economic policy uncertainty and bank stock indices. This limitation restricted the inclusion of more countries in our study. The research relies on monthly data spanning from January 1985 to December 2022, with variations in periods across countries due to the variable ages, particularly in BANK and EPU. This suggests a potential avenue for future research to explore diverse regions globally, providing additional support for our findings. Subsequent studies can also employ alternative data sources and statistical methods to further examine the relationship between the variables under consideration.

As previously stated, the Elevated Policy Uncertainty (EPU) exerts a detrimental influence on banks in the majority of developed economies. Specifically, during crises, a rise in EPU correlates with a decline in bank stock values. Conversely, an escalation in oil prices is associated with an increase in bank stock values amid crises.

The implications of our findings are of significant importance for policy considerations, suggesting the need for comprehensive reforms rather than relying solely on specific macro prudential policies. To prevent or mitigate potential systemic risks, measures should be implemented to reduce banks' vulnerability to both government policy uncertainty and oil price shocks, particularly during stress periods. These reforms should aim to enhance the resilience of banks and reduce their exposure to external shocks, thereby promoting stability and safeguarding the financial system. Bank managers play a crucial role in protecting their institutions from significant losses and economic downturns during crises. It is imperative for them to develop a robust plan that not only safeguards the

bank's financial performance but also mitigates potential losses stemming from borrower defaults. Proactive measures, such as implementing effective risk management strategies, diversifying loan portfolios, and strengthening credit assessment processes, can help banks hedge against defaults and navigate economic and financial downturns. By devising comprehensive plans and strategies, bank managers can safeguard their institutions and support clients in weathering challenging economic conditions. Lastly, bank stocks are appealing assets for investors during stable economic periods. However, during economic crises, they become highly risky and vulnerable to uncertain government policies and oil price fluctuations. To mitigate this systematic risk, investors and portfolio managers should consider implementing investing and hedging strategies. These strategies could include diversifying their investment portfolios, incorporating risk management tools, and closely monitoring the impact of economic uncertainties and oil price movements on bank stocks. By adopting proactive approaches, investors can navigate the challenges posed by such systematic risks and potentially enhance their overall investment outcomes.

Data availability statement

All data used in this study is available upon request.

CRediT authorship contribution statement

Diary Jalal Ali: Writing – original draft, Methodology, Formal analysis. **Boren Sargon:** Writing – review & editing, Supervision, Resources, Methodology, Conceptualization. **Diawar Mahdi Hadi:** Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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